Complex Eigenvalues and the Trace–Determinant Plane

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Question: What do complex eigenvalues mean in a real system?

Let $\alpha \in \mathbb{R}$, $\beta \in \mathbb{R} \setminus \{0\}$, and

$$\begin{pmatrix} \dot{x} \\ \dot{y} \end{pmatrix} = \begin{pmatrix} \alpha & -\beta \\ \beta & \alpha \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix}.$$

Find the eigenvalues and the polar equations.

Solution. The characteristic polynomial is

$$(\alpha - \lambda)^2 + \beta^2,$$

so

$$\lambda = \alpha \pm i\beta.$$

For $r > 0$ and $\theta = \arg(x + iy)$,

$$r\dot{r} = x\dot{x} + y\dot{y} = \alpha(x^2 + y^2) = \alpha r^2,$$

$$\dot{\theta} = \frac{x\dot{y} - y\dot{x}}{r^2} = \beta.$$

Therefore

$$r(t) = r_0 e^{\alpha t}, \qquad \theta(t) = \theta_0 + \beta t.$$

The origin is an asymptotically stable spiral if $\alpha < 0$, an unstable spiral if $\alpha > 0$, and a center if $\alpha = 0$. The angular revolution time is $2\pi/|\beta|$, and the sign of β gives the orientation. When $\alpha \neq 0$, this is the time for one revolution, not a period of the nonclosed trajectory.

Notation: Trace

For

$$A = \begin{pmatrix} a & b \\ c & d \end{pmatrix},$$

write

$$\tau = \operatorname{tr} A = a + d.$$

Notation: Determinant

For $A = \begin{pmatrix} a & b \\ c & d \end{pmatrix}$, write

$$\Delta = \det A = ad - bc.$$

Notation: Discriminant

For a real 2×2 matrix with trace τ and determinant Δ, write

$$D = \tau^2 - 4\Delta.$$

Definition: Matrix exponential

For a real 2×2 matrix A and $t \in \mathbb{R}$, the **matrix exponential** is

$$e^{At} := \sum_{k=0}^{\infty} \frac{(At)^k}{k!}.$$

Observation: Eigenvalues from trace and determinant

The characteristic polynomial and eigenvalues are

$$p(\lambda) = \lambda^2 - \tau\lambda + \Delta, \qquad \lambda_{1,2} = \frac{\tau \pm \sqrt{D}}{2}.$$

Thus

$$\lambda_1 + \lambda_2 = \tau, \qquad \lambda_1\lambda_2 = \Delta.$$

Theorem: Trace–determinant classification for an isolated origin

Let A be a real 2×2 matrix with $\Delta = \det A \neq 0$. Then the origin is the unique equilibrium of $\dot{\mathbf{x}} = A\mathbf{x}$, and

condition	type at the origin	stability
$\Delta < 0$	saddle	unstable
$\Delta > 0,\ D > 0$	node	$\tau < 0:$ asymptotically stable; $\tau > 0:$ unstable
$\Delta > 0,\ D < 0,\ \tau \neq 0$	spiral	$\tau < 0:$ asymptotically stable; $\tau > 0:$ unstable
$\Delta > 0,\ \tau = 0$	center	neutrally stable
$\Delta > 0,\ D = 0$	star or degenerate node	$\tau < 0:$ asymptotically stable; $\tau > 0:$ unstable

Proof. If $\Delta < 0$, the eigenvalues are real with opposite signs. If $\Delta > 0$ and $D > 0$, they are real

Theorem: Trace–determinant classification for an isolated origin (continued)

with the same sign, determined by their sum τ. If $D < 0$, they are

$$\lambda_{1,2} = \frac{\tau}{2} \pm \frac{i}{2}\sqrt{4\Delta - \tau^2},$$

so $\tau/2$ controls radial growth or decay.
If $\tau = 0$ and $\Delta > 0$, Cayley–Hamilton gives

$$A^2 = -\Delta I,$$

and hence

$$e^{At} = I\cos(\sqrt{\Delta}\,t) + \frac{A}{\sqrt{\Delta}}\sin(\sqrt{\Delta}\,t).$$

Every nonzero solution is periodic with period $2\pi/\sqrt{\Delta}$.
If $D = 0$ and $\Delta > 0$, the repeated eigenvalue is $\tau/2 \neq 0$; the eigenspace dimension distinguishes a star from a degenerate node.

Proposition: Zero-determinant boundary

Let A be a real 2×2 matrix with $\Delta = \det A = 0$, and let $\tau = \operatorname{tr} A$.

condition	equilibria	stability of the origin
$\tau < 0$	$\ker A$ is a line	Lyapunov stable, not attracting
$\tau > 0$	$\ker A$ is a line	unstable
$\tau = 0,\ A = 0$	$\mathbb{R}^2$	Lyapunov stable, not attracting
$\tau = 0,\ A \neq 0$	$\ker A$ is a line	unstable

Proof. If $\tau \neq 0$, the distinct eigenvalues are 0 and τ. The zero eigendirection consists of equilibria, while the other component decays for $\tau < 0$ and grows for $\tau > 0$.
If $\tau = 0$, Cayley–Hamilton gives $A^2 = 0$. When $A = 0$, every solution is constant. When $A \neq 0$,

$$e^{At} = I + tA,$$

so arbitrarily small initial states outside $\ker A$ produce solutions that leave any fixed neighborhood of the origin.

Visualization: Trace–determinant plane

With horizontal coordinate τ and vertical coordinate Δ,

region or curve	portrait
$\Delta < 0$	saddle
$0 < \Delta < \tau^2/4$	node
$\Delta > \tau^2/4,\ \tau < 0$	asymptotically stable spiral
$\Delta > \tau^2/4,\ \tau > 0$	unstable spiral
$\tau = 0,\ \Delta > 0$	center
$\Delta = \tau^2/4 > 0$	star or degenerate node
$\Delta = 0,\ \tau < 0$	Lyapunov-stable line of equilibria
$\Delta = 0,\ \tau > 0$	unstable line of equilibria
$\Delta = \tau = 0$	zero matrix or unstable nilpotent boundary

Visualization: Trace–determinant plane (continued)

Within the half-plane $\Delta > 0$, the parabola $\Delta = \tau^2/4$ separates real-eigenvalue nodes from complex-eigenvalue portraits; the latter are spirals except for centers on $\tau = 0$. If $\tau < 0$ and $\Delta > 0$, the origin is asymptotically stable; if $\tau < 0$ and $\Delta = 0$, it is Lyapunov stable but not attracting. If $\tau > 0$, the origin is unstable, and every point with $\Delta < 0$ is a saddle.

Question: Can trace and determinant classify without solving?

Classify

$$A_1 = \begin{pmatrix} 1 & 2 \\ 3 & 4 \end{pmatrix}, \qquad A_2 = \begin{pmatrix} 2 & 1 \\ 3 & 4 \end{pmatrix}.$$

Solution. For A_1,

$$\tau = 5, \qquad \Delta = 4 - 6 = -2 < 0,$$

so the origin is a saddle.
For A_2,

$$\tau = 6, \qquad \Delta = 8 - 3 = 5, \qquad D = 36 - 20 = 16 > 0.$$

The origin is an unstable node. Its eigenvalues are

$$\lambda = \frac{6 \pm 4}{2} = 5, 1.$$

Question: What is the complete solution of an asymptotically stable spiral?

For

$$A = \begin{pmatrix} -1 & -4 \\ 1 & -1 \end{pmatrix},$$

classify the origin, determine the orientation, and solve the initial-value problem $\mathbf{x}(0) = (2, 0)$.

Solution.

$$\tau = -2, \qquad \Delta = 5, \qquad D = -16,$$

so $\lambda = -1 \pm 2i$ and the origin is an asymptotically stable spiral. At $(1, 0)$,

$$A \begin{pmatrix} 1 \\ 0 \end{pmatrix} = \begin{pmatrix} -1 \\ 1 \end{pmatrix},$$

so the rotation is counterclockwise.
Let

$$B = A + I = \begin{pmatrix} 0 & -4 \\ 1 & 0 \end{pmatrix}.$$

Since $B^2 = -4I$,

$$e^{At} = e^{-t}\left(I \cos 2t + \frac{B}{2} \sin 2t\right) = e^{-t} \begin{pmatrix} \cos 2t & -2 \sin 2t \\ \frac{1}{2} \sin 2t & \cos 2t \end{pmatrix}.$$

Thus

$$x(t) = 2e^{-t} \cos 2t, \qquad y(t) = e^{-t} \sin 2t.$$

One revolution takes time π, and the radial envelope decays by $e^{-\pi}$ per turn. The spiral trajectory itself is not periodic.

Question: Why do boundary portraits require more than eigenvalue multiplicity?

Classify

$$A_1 = -I, \qquad A_2 = \begin{pmatrix} -1 & 1 \\ 0 & -1 \end{pmatrix}, \qquad A_3 = \begin{pmatrix} -1 & 0 \\ 0 & 0 \end{pmatrix}.$$

Solution. Both A_1 and A_2 satisfy

$$\tau = -2, \qquad \Delta = 1, \qquad D = 0.$$

For A_1, every nonzero vector is an eigenvector, so the origin is an asymptotically stable star node.
For A_2,

$$\ker(A_2 + I) = \operatorname{span}\{(1, 0)\},$$

so the origin is an asymptotically stable degenerate node.
For A_3, $\Delta = 0$, $\tau = -1$, and

$$x(t) = x_0 e^{-t}, \qquad y(t) = y_0.$$

Every point on the y-axis is an equilibrium. The origin is Lyapunov stable but not attracting.

Definition: Linear love-affair model

Let $R(t)$ denote Romeo's feeling toward Juliet and $J(t)$ Juliet's feeling toward Romeo, with positive values representing love and negative values representing hate. A linear response model is

$$\dot{R} = aR + bJ, \qquad \dot{J} = cR + dJ.$$

This is the **linear love-affair model**.

Observation: Interpretation of the response coefficients

In the linear love-affair model, the diagonal coefficients a, d measure responses to one's own feelings, while the off-diagonal coefficients b, c measure responses to the other person's feelings.

Question: What happens to responsive Romeo and fickle Juliet?

Let

$$\dot{R} = aJ, \qquad \dot{J} = -bR, \qquad a, b > 0.$$

Classify the origin and find the trajectories and period.

Solution.

$$A = \begin{pmatrix} 0 & a \\ -b & 0 \end{pmatrix}, \qquad \tau = 0, \qquad \Delta = ab > 0.$$

Thus the origin is a center with eigenvalues $\pm i\sqrt{ab}$. Moreover,

$$H(R, J) = bR^2 + aJ^2$$

is conserved because

$$\dot{H} = 2bR(aJ) + 2aJ(-bR) = 0.$$

For $C > 0$, the level sets $H = C$ are the nonzero trajectories and are ellipses. The level set $H = 0$ is the equilibrium at the origin. Every nonzero trajectory has period

$$T = \frac{2\pi}{\sqrt{ab}}.$$

Question: What happens to responsive Romeo and fickle Juliet? (continued)

At $(R, 0)$ with $R > 0$, $\dot{J} = -bR < 0$, so the motion is clockwise.

Question: What happens to two identically cautious lovers?

Let

$$\dot{R} = aR + bJ, \qquad \dot{J} = bR + aJ, \qquad a < 0 < b.$$

Give the complete classification as b varies.

Solution. The eigenpairs are

$$\lambda_+ = a + b, \quad \mathbf{v}_+ = (1, 1), \qquad \lambda_- = a - b, \quad \mathbf{v}_- = (1, -1).$$

Since $a - b < 0$,

condition	type	long-time behavior
$0 < b < -a$	asymptotically stable node	$(R, J) \to (0, 0)$
$b = -a$	line $R = J$ of equilibria	$(R, J) \to (c, c)$
$b > -a$	saddle	$W^u = \{R = J\},\ W^s = \{R = -J\}$

In the middle case, $c = (R(0) + J(0))/2$.